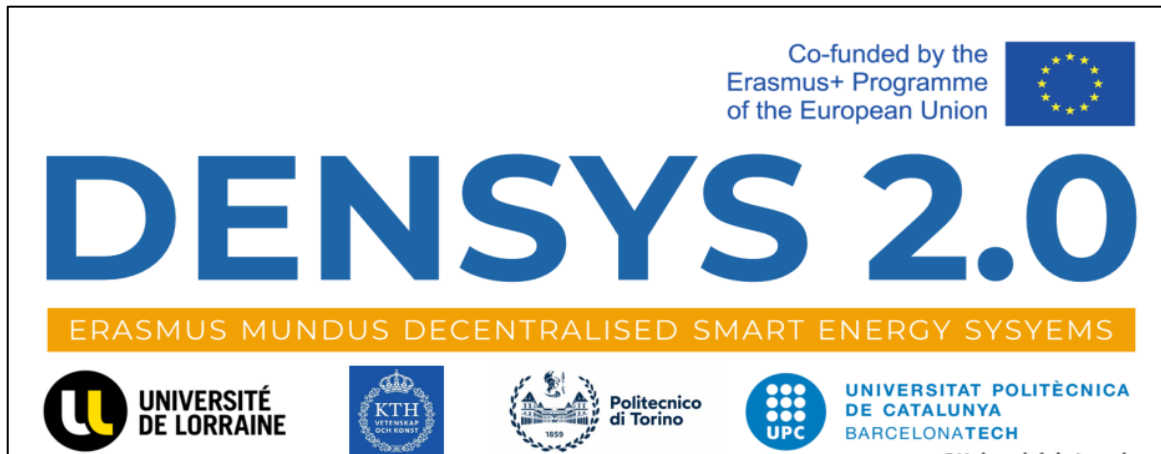




DATA AND FORECASTING IN MICROGRIDS

SECONDARY CONTROL: A RESIDENTIAL MICROGRID

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INTRODUCTION

Our project is a residential microgrid which has photovoltaic panels, a battery storage system, a diesel generator, a heat pump, household load, an electric vehicle and grid connection for importing and exporting electricity. Our objective is to satisfy all the electrical loads, maintain thermal comfort, charge the EV and battery within all the required operating limits.

General assumptions:

- 1. Yearly load:** The yearly load energy was fixed at 10000kWh for a 100m² house with washing machine, dryer, dishwasher, freezer which matches the total annual household consumption for a house with multiple occupants in France given by [ENGIE, 2026](#).
- 2. EV usage:** According to [RTE, 2019](#), EVs in France travel up to 12,500km per year and we have used this value as the yearly EV distance for the optimization problem.
- 3. Comfort range:** A temperature range of $\pm 2^{\circ}\text{C}$ from the set temperature was assumed.

OPERATIONAL PLANNING

After implementing each device constraint and minimizing the objective function i.e. the cost of running the system, an optimal solution with all constraints satisfied was found as shown in Table 1 below. The results show that the available PV was maximally used, in conjunction with the battery, EV, and imported power (as a last resort) to satisfy all load demands. There was no need for generator usage for the base case.

The optimization across 365 days showed noticeable peaks in the system caused majorly by the EV attempting to reach the target value in a short period of time and the Heat pump attempting to reach set temperature range; these resulted in spikes from the grid importing a large amount of power from the grid to meet the load demand as shown in Figures 1,3 and 5.

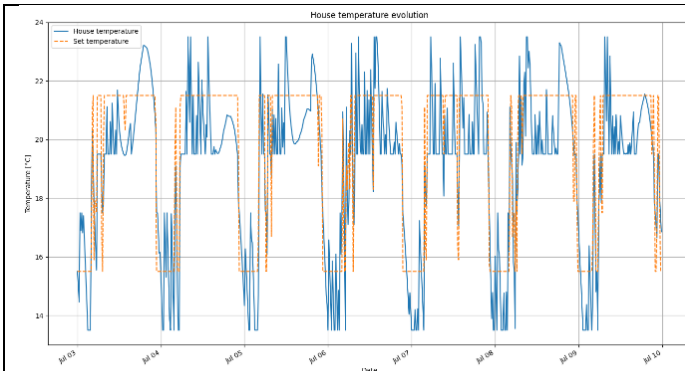


Figure 1: Summer with no peak penalty

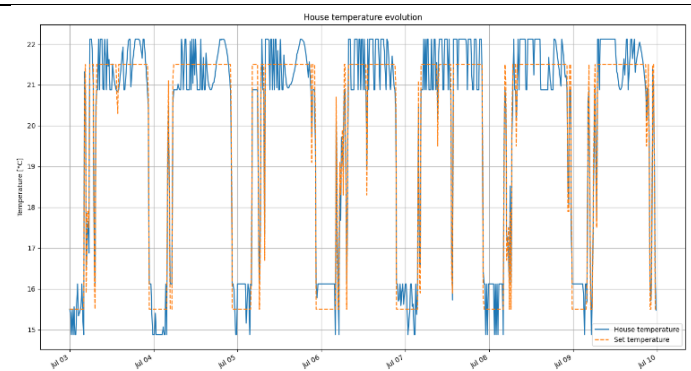


Figure 2: Summer with peak penalty

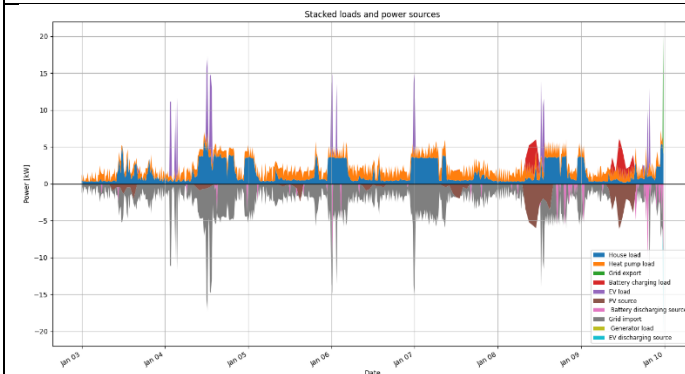


Figure 3: Winter power balance with no peak penalty

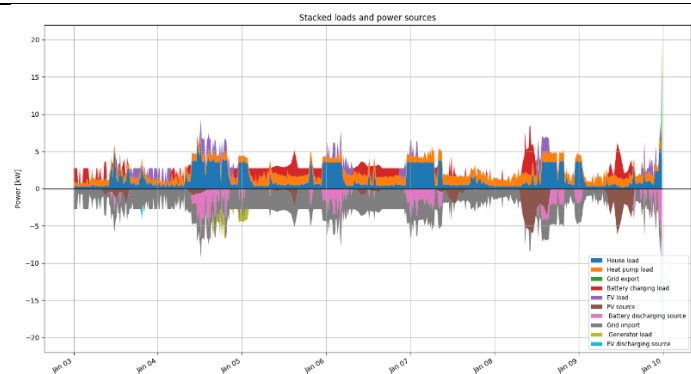
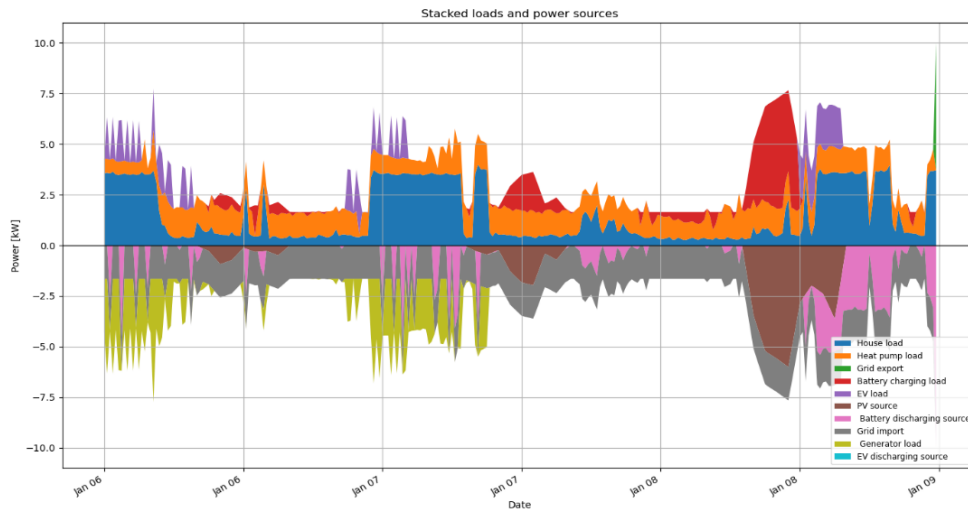
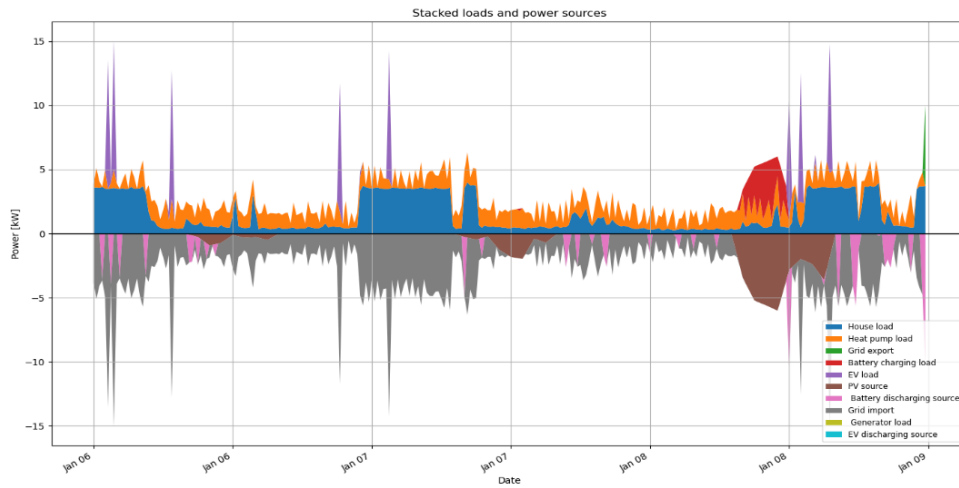
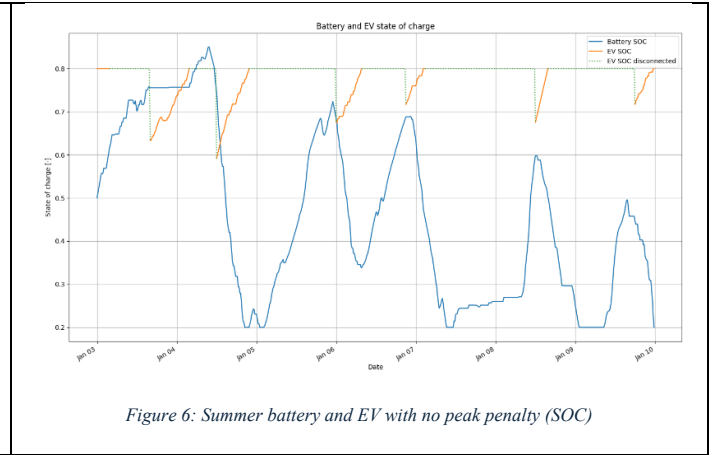
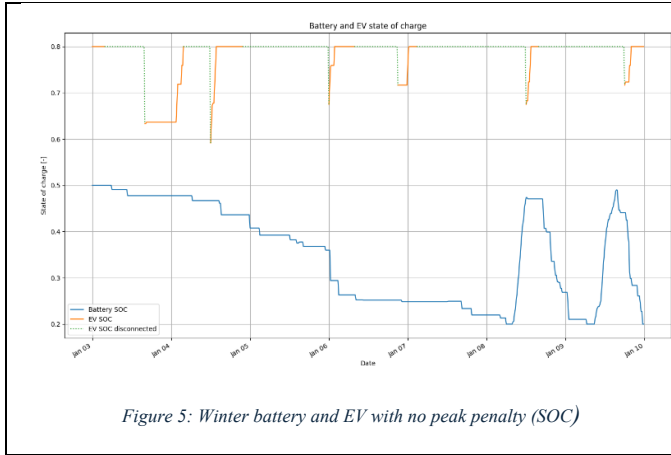


Figure 4: Winter power balance with peak penalty

To resolve these, we included peak penalties for the import, EV and temperature deviations to normalize the peaks as shown in Figures 2, 4 and 6. The resulting differences in the system are shown in Table 1.



The figures 7 and 8 above show a comparison of the load balance before and after including the peak penalty for 3 days (Jan 06- Jan 09) in the operational planning model. The generator is seen to turn on briefly in the peak-penalty case. This is because the optimizer may prefer using the generator for short periods to avoid

large grid import peaks or high battery discharge. Even though generator energy is more expensive than grid import, the peak penalty makes reducing extreme power peaks more important.

From our optimization, the PV, battery and grid import were sufficient to fulfil all the load demand of the house without running the generator. All the PV energy available was used and some revenue was generated during the year from the electricity exported to the grid. The cost of operation was estimated to be **2885.83 EUR/year** for the system without peak penalties and **2893.05 EUR/year** for the system with peak penalties.

Table 1: Yearly system costs with and without peak penalty

	Yearly system costs no peak penalty	Yearly system costs -peak penalty
HP energy	5139.44 kWh	5139.44 kWh
EV charging energy	2642.63 kWh	2614.10 kWh
EV discharging energy	110.03 kWh	82.63 kWh
PV energy used	14305.98 kWh	14305.98 kWh
PV available	14305.98 kWh	14305.98 kWh
BSS charging energy	4146.68 kWh	4271.54 kWh
BSS discharging energy	3756.28 kWh	3868.96 kWh
Grid Imported Energy	7494.97 kWh	7513.57 kWh
Grid Exported Energy	3738.50 kWh	3746.07 kWh
Generator energy	00.00 kWh	00.00 kWh
Generator cost	0.00 EUR/year	0.00 EUR/year
Grid import cost	2997.99 EUR/year	3005.43 EUR/year
Grid export revenue	112.16 EUR/year	112.38 EUR/year
Net yearly OPEX	2885.83 EUR/year	2893.05 EUR/year

Figure 9 shows the cumulative cost breakdown over the full year. The cumulative net cost increases gradually and reaches about 2,900 EUR by the end of the year. The net cost follows the grid import cost closely, which shows that grid import is the main contributor to the yearly operating cost. The generator cost remains zero throughout the year, meaning the optimizer did not need to use the generator under the selected price assumptions. The export revenue increases slowly and only offsets a small part of the import cost. The sharper cost increase at the beginning and end of the year is due to higher winter demand, especially from the heat pump, combined with lower PV availability. During periods with better PV generation, the cost curve increases more slowly because part of the demand is covered by PV and battery storage.

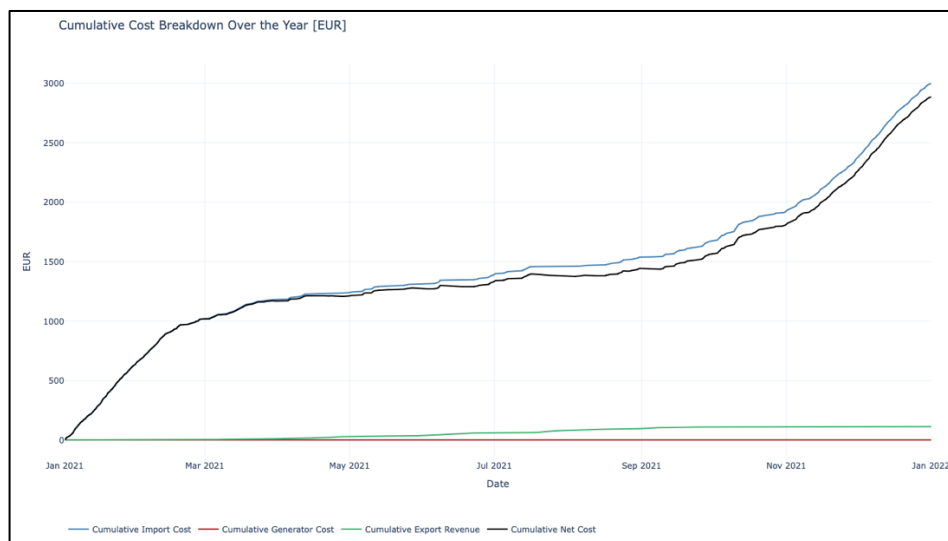


Figure 9: Cumulative cost breakdown over the year for the operational planning case

Figure 10a shows the effect of evolving energy(import) prices. The import price was varied from 0.2 to 0.8 EUR/kWh while maintaining generator and export prices. The import cost increases until it reaches 0.6 EUR/kWh (generator price), then the optimiser finds the generator to be cheaper, and so runs the generator instead. The net yearly OPEX increases and remains constant from this point. In Figure 10b, the export price was also varied, and as expected, the export revenue increases while the net yearly OPEX reduces. Figure 10c shows the variation of the generator price. With increasing generator price, the system runs the generator and so the cost increases until the price reaches 0.4 EUR/kWh (import price), then the optimiser finds the importing from the grid to be cheaper. The net yearly OPEX increases and remains constant from this point.

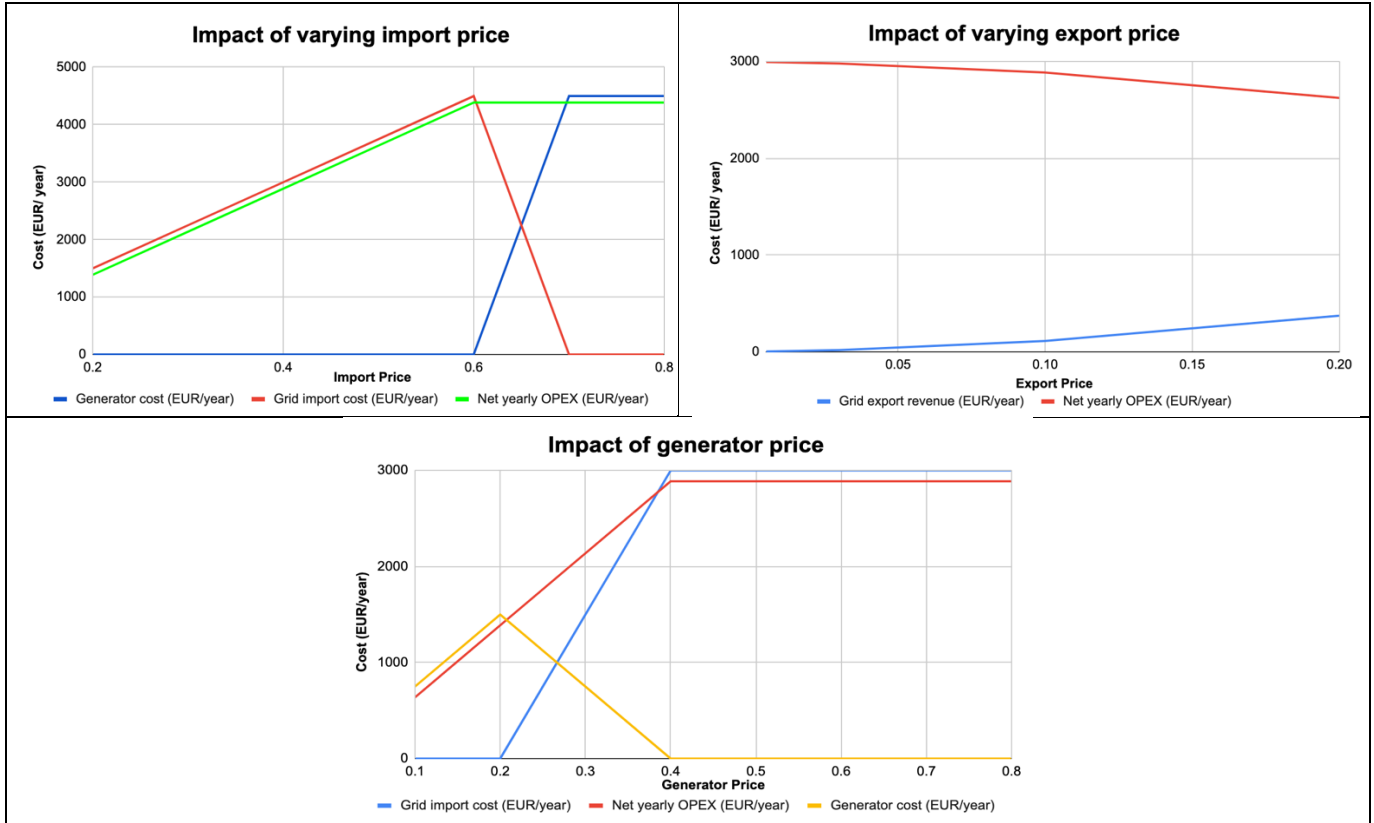


Figure 10: Impact of evolving energy prices on the system

SYSTEM SIZING

For the base case of the system sizing, additional constraints were included for the PV inverter size and battery inverter size. Using a discount factor of 8%, the base case gave an NPV of OPEX of 37,181.89 EUR, a CAPEX of 20,504.38 EUR, and a total NPV cost of 57,686.27 EUR. This makes the base case the most economical option because the system can still use grid import when needed and export excess PV energy for revenue.

Compared with the base case, the isolated microgrid has a much higher total NPV cost of 75,995.80 EUR because the system has no access to grid import or export. As a result, it requires a much larger battery capacity of 34.59 kWh. The system also uses a lot of generator energy of 7297.85 kWh during the year to fulfil the demand when PV generation or battery storage is not sufficient, which increases the yearly OPEX.

The export-not-paid case has a slightly higher total cost than the base case at 58,515.21 EUR. Even though this case is still connected to the grid, the excess PV energy exported to the grid has no financial value, reducing the economic benefit of installing and exporting PV power. The results for comparing all cases are detailed in Table 2 below.

Table 2: Comparison Table for System sizing on Base Case, Isolated Microgrid and Export-not-paid case.

	Base Case	Isolated microgrid	Export not paid
PI Gen (EUR/kWh)	0.6	0.6	0.6
PI Import (EUR/kWh)	0.4	0.4	0.4
PI Export (EUR/kWh)	0.03	0.03	0
Fixed load energy (kWh)	10000	10000	10000
HP energy (kWh)	5139.44	6773.25	5247.91
EV charging energy (kWh)	2583.93	3075.22	2612.84
EV discharging energy (kWh)	53.65	340.25	49.14
PV energy used (kWh)	11484.8	12778.37	10010.06
PV available (kWh)	11630.47	16206.59	10191.31
BSS charging energy (kWh)	2257.56	5491	2378.94
BSS discharging energy (kWh)	2042.89	4923.01	2146.36
Grid Imported Energy (kWh)	9716.41	0	10220.44
Grid Exported Energy (kWh)	3316.82	0	2186.3
Generator energy (kWh)	0	7297.85	0
PV Capacity (kWp)	8.13	1.33	7.12
PV Inverter (kW)	6.02	7.11	5.13
Battery Capacity (kWh)	15.65	34.59	15.07
Battery Inverter (kW)	3.91	8.65	3.77
Generator Max Power (kW)	0	3.76	0
Generator cost (EUR/year)	0	4378.71	0
Grid import cost (EUR/year)	3886.56	0	4088.18
Grid export revenue (EUR/year)	99.5	0	0
Yearly OPEX (EUR/year)	3787.06	4378.71	4088.18
CAPEX (EUR)	20504.38	33004.98	18376.88
NPV of OPEX (EUR)	37181.89	42990.82	40138.33
Total NPV cost (EUR)	57686.27	75995.8	58515.21

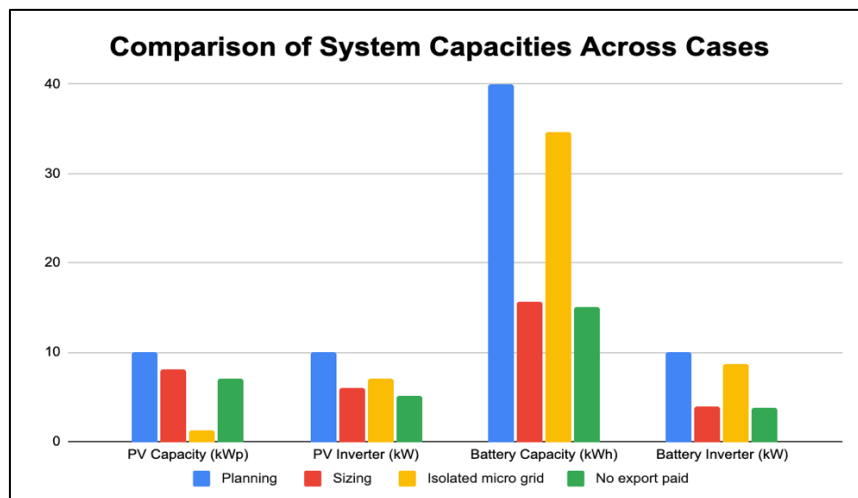


Figure 11: Comparison of capacities

From the comparison between battery and PV capacities for the planning case, optimized sizing case, isolated microgrid, and no-export-paid case shown in Figure 11, the planning case uses the already defined sizes, so it has the largest battery capacity at 40 kWh and fixed inverter capacities of 10 kW. After optimization, the

sizing case reduces the PV, battery, and inverter capacities because the model chooses only the capacity that was needed for the load demand for the year and hence minimize the total cost.

The isolated microgrid requires a much larger battery capacity and battery inverter than the grid-connected sizing case because it cannot rely on grid import when PV is not available. This makes battery storage more important for reliability.

The no-export-paid case has a lower PV and battery size than the base sizing case because excess PV energy has no financial value when exported. Therefore, the optimizer avoids oversizing the PV system.

Result of varying the CAPEX on the Investment

While varying the CAPEX budget, an increase in the budget shows a decrease in the NPV of OPEX (Cash Flow) because the increase leads to higher investment in more PV and battery capacity, which reduces dependence on grid import and in turn the net yearly OPEX.

The blue and red markers shows the position of the NPV of OPEX and the Total NPV Cost in the base case and the trend of the increase of NPV with an increase in CAPEX and thus a reduction in OPEX.

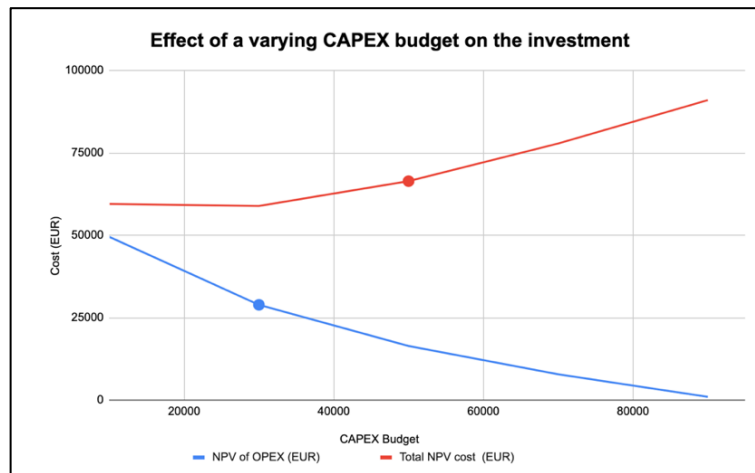


Figure 12: Effect of a varied CAPEX budget on the investment

Result of varying the Load

Varying the fixed house load from 5000 kWh to 14000 kWh, results in an increase in the required PV capacity, battery capacity, and grid imported energy because the original capacities are not sufficient to meet the increase in the load demand.

The result shows that the fixed household load has a direct effect on system sizing. A higher load would require larger generation and storage capacity, but it also increases grid dependence. With an increase in the capacities of this system, there will be a corresponding increase in the CAPEX, NPV and the yearly OPEX of the system.

Table 3: Effect of load variation on the system sizing

Load variation	PV Capacity (kWp)	Battery Capacity (kWh)	Grid Imported Energy (kWh)
5000	5.7	11.51	7065.94
8000	7.08	14.1	8670.83
10000	8.13	15.65	9716.41
12000	8.97	17.2	10858.67
14000	9.8	18.48	12039.49

Varying the distances travelled by the EV

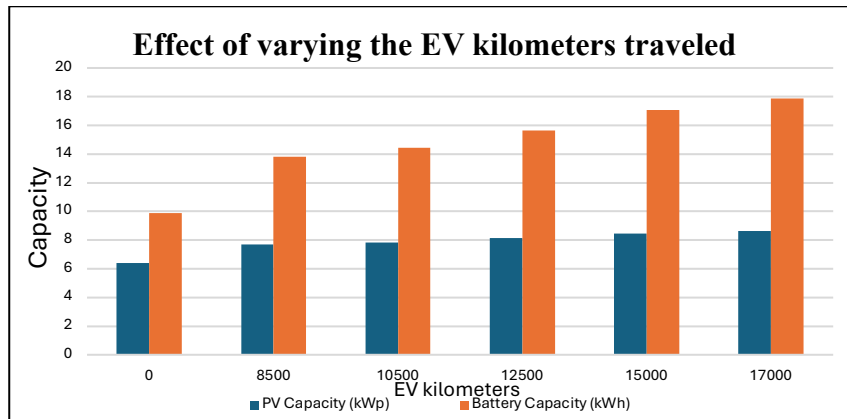


Figure 13: Effect of a varying EV kilometers travelled on the system

The effect of varying the yearly EV kilometres travelled was also studied. As the EV travels for longer distances, the charging energy required increases, which creates more demand on the microgrid. This leads to increase in both the PV and battery sizing as shown in Figure 13 above.

Result of varying the energy costs during investment horizon

Import cost sensitivity: As the system cost changes over the 20-year investment horizon, the grid import cost increases steadily as the import price rises, reaching its highest value at around 0.6 EUR/kWh as shown in Figure 14a. After this point, the grid import cost drops sharply to almost zero, while the generator cost increases suddenly and remains constant. This shows that once grid electricity becomes too expensive, the optimizer stops relying on grid import and switches to the generator as the cheaper supply option. The grid export revenue remains very small compared with the import and generator costs.

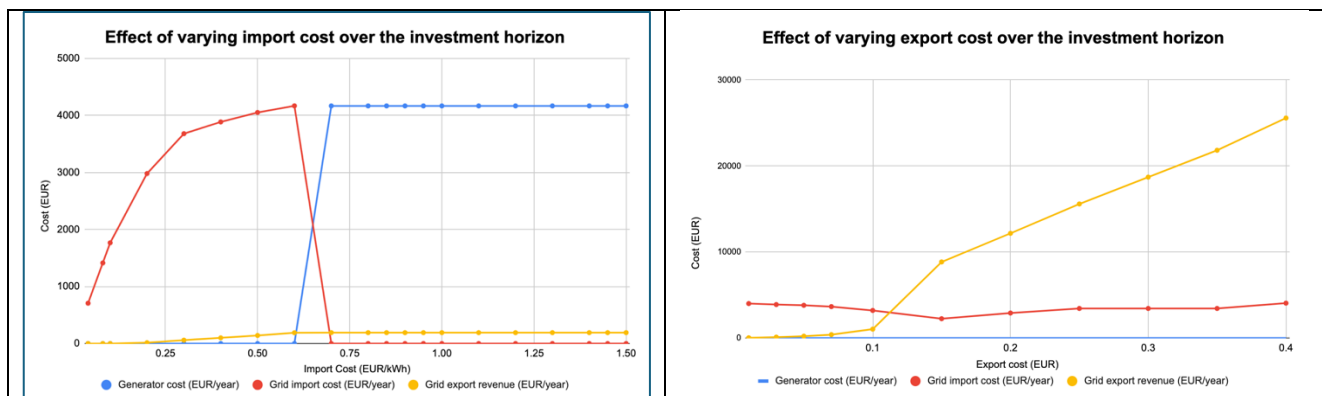


Figure 14: Impact of evolving energy prices on the system

Export cost sensitivity: Figure 14b shows the effect of varying the export price over the 20-year investment horizon. The grid export revenue increases steadily as the export price rises. This also affects the optimal system size because higher export prices make it more attractive to install larger PV capacity and PV inverter capacity, since excess PV can now generate revenue instead of being wasted or sold at a low price. At the same time, the yearly OPEX decreases because the revenue from exported electricity offsets part of the investment and operating cost. Eventually, the revenue from exports offsets the total NPV costs and the system starts to make profit.

The model becomes unbounded when the export price becomes equal to the import price because there is no longer a penalty for buying and selling electricity at the same value. If the optimization model does not prevent simultaneous grid import and export, the optimizer can create an artificial loop where it imports electricity and exports the same amount back to the grid.

CO₂ Emissions Modelling

CO₂ emissions were included in the sizing optimization by considering both operational and embodied emissions. We defined a penalty for the CO₂ emissions per gram of CO₂ emitted to be 0.000075 EUR/g (benchmark value according to European Carbon Border Adjustment Mechanism (CBAM)). We then modified the objective function so that the system minimizes the total cost CO₂ emissions included. In addition, a maximum annual emissions limit was also introduced. The total annual CO₂ emissions were constrained to remain below 3 tonnes/year to ensure that the optimized system is not only cost-effective but also satisfies a fixed environmental target.

Operational emissions come from generator use and grid imports, while the embodied emissions come from the installed equipment, including PV system, batteries, inverters, and the generator over the 20-year period. A CO₂ price was then applied to convert emissions into a monetary penalty. Operational CO₂ was included in the OPEX, while embodied CO₂ was included in the CAPEX. Therefore, the optimization selected a system that not only reduces cost and considers the environmental impact of the selected technologies but also remains below the annual CO₂ emissions limit.

Table 4: CO₂ emissions value

CO ₂ component	Value
Operational CO ₂	1.09 tonnes CO ₂ /year
Embodied CO ₂ over 20 years	38.23 tonnes CO ₂
Annualized embodied CO ₂	1.91 tonnes CO ₂ /year
Total annualized CO ₂	3.00 tonnes CO ₂ /year

For the optimized CO₂ case, the total annualized CO₂ emissions were 3 tonnes CO₂/year. The operational emissions were 1.09 tonnes CO₂/year, while the embodied emissions were 38.23 tonnes CO₂ over the 20-year investment horizon, equivalent to 1.91 tonnes CO₂/year when annualized.

Table 5: Cost breakdown as an effect of CO₂ emissions

Cost component	Value
CAPEX	49,923.81 EUR
NPV of OPEX over 20 years	74,239.98 EUR
Total NPV cost	124,163.79 EUR
Net yearly OPEX	7,561.51 EUR /year

In terms of cost, the optimized CO₂ case gives a total NPV cost of 124,163.79 EUR over the 20-year investment horizon. This consists of 49,923.81 EUR in CAPEX and 74,239.98 EUR as the NPV of OPEX. The net yearly OPEX is 7,561.51 EUR/year, after considering grid import cost and export revenue. Since the generator is not used, the operating cost is mainly driven by grid import, while the CAPEX is mainly driven by the selected PV and battery sizes.

APPENDIX

Link to Google sheets: [Data and sensitivity sheet](#)

Mathematical Modelling (Planning):

Objective function: $\min(\sum(\pi_{gen}P_{gen}(t) + \pi_{import}P_{import}(t) + \pi_{export}P_{export}(t)) \times dt)$

The objective function is subject to the following constraints:

Power balance: $P_{PV}(t) + P_{gen}(t) + P_{imp}(t) + P_{dis}^{BSS}(t) + P_{dis}^{EV}(t) = P_{load}(t) + P_{HP_heating}(t) + P_{HP_cooling}(t) + P_{ch}^{BSS}(t) + P_{exp}(t) + P_{ch}^{EV}(t)$

PV generation: $P_{PV} \leq P_{PV,max} \times C_{PV}$ & $P_{PV} \leq P_{PV,nom}$

The Generator: $P_{gen} \leq P_{gen}^{max} \cdot u_{gen}$

Battery state of charge: $SOC_{BSS}^{next} = SOC_{BSS} + \frac{\Delta t}{C_{BSS}} \left(\eta_{BSS} \cdot P_{ch}^{BSS} - \frac{P_{dis}^{BSS}}{\eta_{BSS}} \right)$
 $SOC_{BSS}^{min} \times C_{BSS} \leq SOC_{BSS} \leq SOC_{BSS}^{max} \times C_{BSS}$ & $P_{ch}^{BSS} \leq P_{BSS,nom}$ & $P_{dis}^{BSS} \leq P_{BSS,nom}$

The EV state of charge: $SOC_{EV}^{next} = SOC_{EV} + \frac{\Delta t}{C_{EV}} \left(\eta_{EV} \cdot P_{ch}^{EV} - \frac{P_{dis}^{EV}}{\eta_{EV}} \right)$

$SOC_{EV}^{min} \times C_{EV} \leq SOC_{EV} \leq SOC_{EV}^{max} \times C_{EV}$
 $EV_{CONNECTED} \times P_{ch}^{EV} \leq P_{EV,nom}$ & $P_{dis}^{BSS} \leq EV_{CONNECTED} \times P_{EV,nom}$

$SOC_{EV}(t-1) \geq SOC_{target,EV} \cdot C_{EV}$ ---- departure target

$P_{dis}^{EV}(t) = 0$ --- No discharge immediately before departure

The heat pump: $T_{HP}^{next} = T_{HP} + \frac{\Delta t}{C_{HP}} (COP_{HP} (P_{HP_heating} - P_{HP_cooling}) - P_{loss})$

$T_{set} - \Delta T_{max} \leq T_{HP}^{next} \leq T_{set} + \Delta T_{max}$ & $P_{HP_heating} \leq P_{HP,max}$ & $P_{HP_cooling} \leq P_{HP,max}$

Mathematical Modelling (Sizing)

The same constraints from the Planning were implemented for the sizing. However, the objective function changed to minimizing the NPV, where:

$$NPV = CAPEX + \frac{OPEX}{(1 + discount\ rate)^{investment\ horizon}}$$

$OPEX = (\sum(\pi_{gen}P_{gen}(t) + \pi_{import}P_{import}(t) + \pi_{export}P_{export}(t)) \times dt)$

$CAPEX = (\pi_{c_bss}P_{c_bss} + \pi_{c_pv}P_{c_pv} + \pi_{c_gen}P_{c_gen} + \pi_{c_inv}(P_{nom_bss} + P_{nom_pv}))$

Other new constraints that were added are:

$$P_{PV,nom} \leq C_{PV}$$

$$0.25 \times C_{BSS} \leq P_{BSS,nom} \leq C_{BSS}$$

Nomenclature

J : Total penalty cost to be minimized.

π_{gen} : Penalty weight for generator usage.

π_{import} : Penalty weight for imported power

π_{export} : Penalty weight for exported power

π_{c_bss} : Cost of Battery capacity

π_{c_pv} : Cost of PV capacity

π_{c_inv} : Cost of Inverter capacity

π_{c_gen} : Cost of Generator capacity

P_{PV} : Actual PV power utilized.

$P_{PV,max}$: Maximum available PV power at the current time step.
 $P_{PV,nom}$: Nominal capacity/limit of the PV system.
 P_{gen} : Power generated by the backup generator.
 P_{gen}^{max} : Maximum power output limits of the generator.
 P_{load} : The fixed electrical demand (load) of the system.
 SOC_{BSS} : Current State of Charge of the battery.
 SOC_{BSS}^{next} : Predicted State of Charge for the next time step.
 P_{ch}^{BSS} : Power used to charge the battery.
 P_{dis}^{BSS} : Power discharged from the battery.
 C_{BSS} : Total energy capacity of the battery.
 η_{BSS} : Efficiency of the battery (charging/discharging).
 SOC_{BSS}^{min} & SOC_{BSS}^{max} : Minimum and maximum allowable State of Charge limits.
 SOC_{EV} : Current State of Charge of the EV.
 SOC_{EV}^{next} : Predicted State of Charge for the next time step.
 P_{ch}^{EV} : Power used to charge the EV.
 P_{dis}^{EV} : Power used to discharge the EV.
 C_{EV} : Total energy capacity of the EV battery.
 η_{EV} : Charging efficiency of the EV.
 $EV_{connected}$: Binary parameter (1 if the vehicle is plugged in, 0 otherwise).
 SOC_{EV}^{min} & SOC_{EV}^{max} : Operational State of Charge limits for the EV.
 T_{HP} : Current temperature of the thermal system (e.g., water tank).
 T_{HP}^{next} : Predicted temperature for the next time step.
 P_{HP} : Electrical power consumed by the heat pump.
 $P_{HP,heating}$: Electrical Power consumed by the heat pump for heating
 $P_{HP,cooling}$: Electrical Power consumed by the heat pump for cooling
 $P_{HP,max}$: Maximum power rating of the heat pump.
 COP_{HP} : Coefficient of Performance of the heat pump.
 C_{HP} : Thermal heat capacity of the system.
 P_{loss} : Thermal power losses to the environment.
 T_{set} : Target temperature setpoint.
 ΔT_{max} : Maximum allowable temperature deviation from the setpoint.
 Δt : Time step duration (discretization interval).